The mental block was broken by Maarten Schmidt, a thirty-two-year-old Dutch astronomer who had recently joined the Caltech faculty. For months he had struggled to understand a spectrum he had taken of 3C273, one of the weird objects. On 5 February 1963, as he sat in his Caltech office carefully sketching the spectrum for inclusion in a manuscript he was writing, the answer suddenly hit him. The four brightest lines in the spectrum were the four standard "Balmer lines" produced by hydrogen gas —the most famous of all spectral lines, the first lines that college physics students learned about in their courses on quantum mechanics. However, these four lines did not have their usual wavelengths. Each was shifted to the red by 16 percent. 3C273 must be an object containing a massive amount of hydrogen gas and moving away from the Earth at 16 percent of the speed of light—enormously faster than any star that any astronomer had ever seen.

Schmidt flew out into the hall, ran into Greenstein, and excitedly described his discovery. Greenstein turned, headed back to his office, pulled out his spectrum of 3C48, and stared at it for a while. Balmer lines were not present at any redshift; but lines emitted by magnesium, oxygen, and neon were there staring him in the face, and they had a redshift of 37 percent. 3C48 was, at least in part, a massive amount of gas containing magnesium, oxygen, and neon, and moving away from Earth at about 37 percent of the speed of light.

What was producing these high speeds? If, as everyone had thought, these weird objects (which would later be named *quasars)* were some type of star in our own Milky Way galaxy, then they must have been ejected from somewhere, perhaps the Milky Way's central nucleus, with enormous force. This was too incredible to believe, and a close examination of the quasars' spectra made it seem extremely unlikely. The only reasonable alternative, Greenstein and Schmidt argued (correctly), was that these quasars are very far away in our Universe, and move away from Earth at high speed as a result of the Universe's expansion.